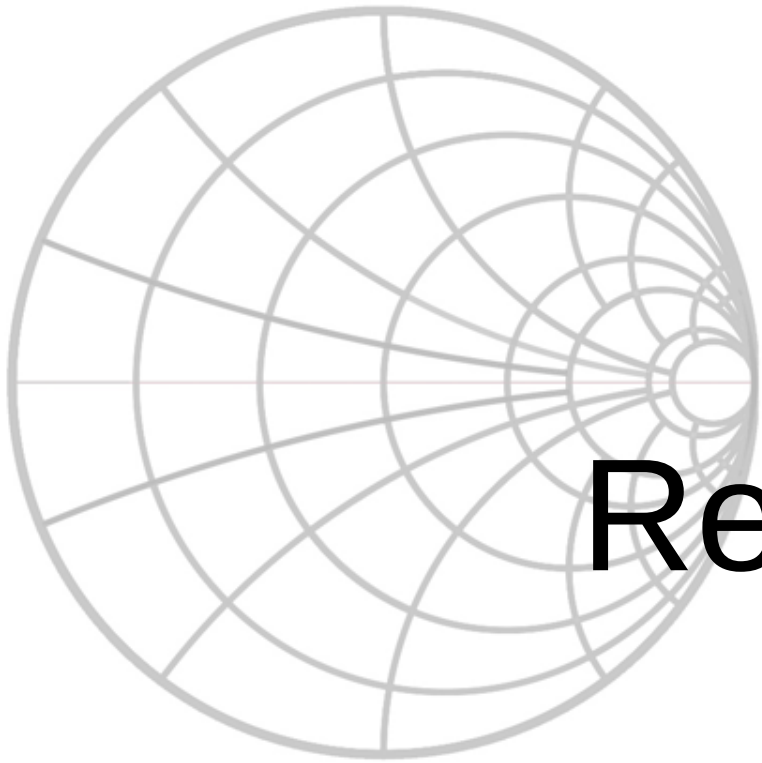




Review Bilangan Kompleks

Muhammad Syirajuddin S.

Penulisan dalam Rectangular

$$z = x + jy$$

$$\Re\{z\} = x \quad \Im(z) = y$$

Contoh :



$$z = 5 + j3$$

$$z = 5 + 3i$$

Penulisan dalam Rectangular

Contoh :

$$z = 5 + j3$$

$$z = 5 + 3i$$

$$\Re\{z\} = 5 \quad \Im\{z\} = 3$$



Penulisan dalam Polar

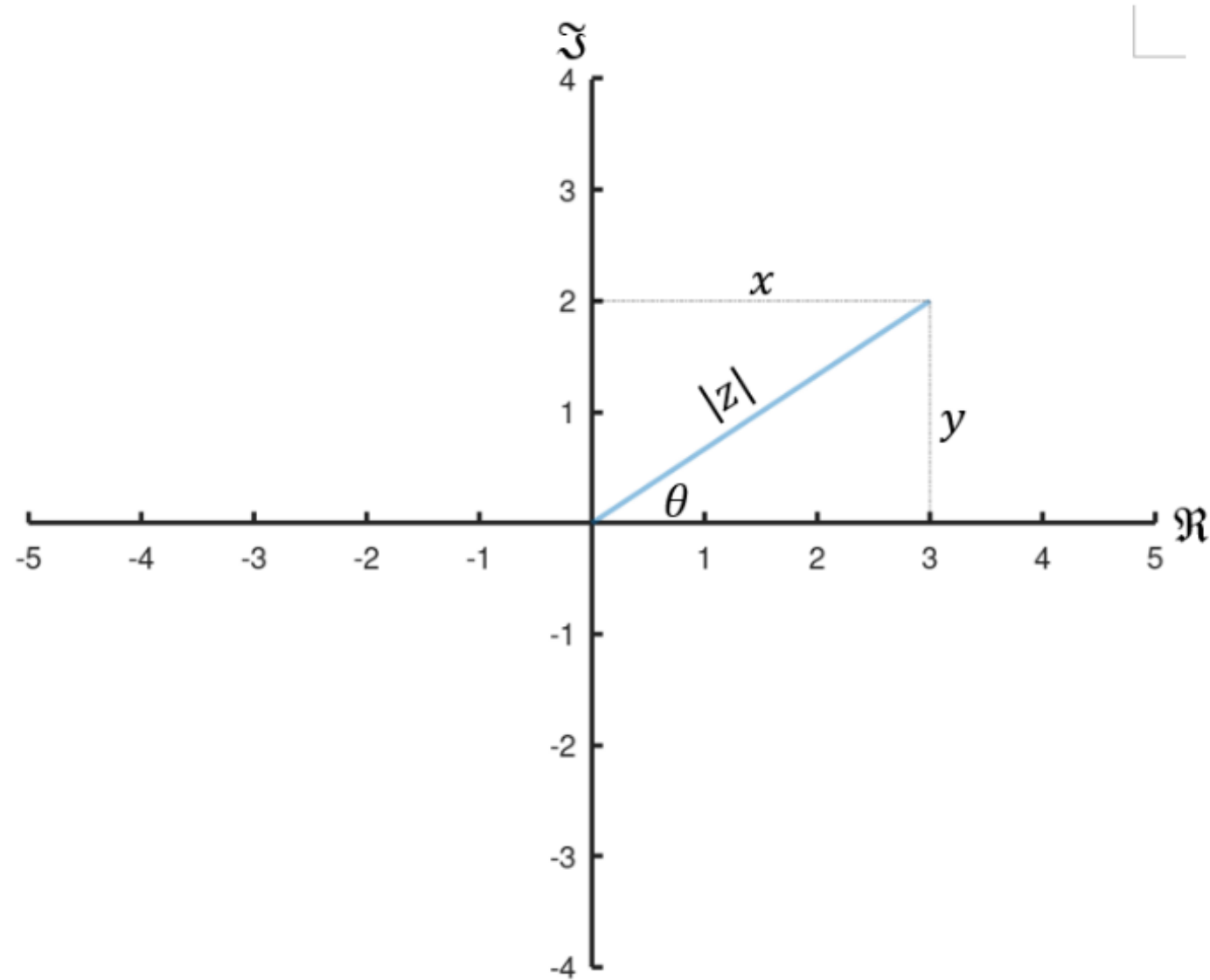
$$z = |z| \angle \theta = |z| e^{j\theta}$$

Contoh :

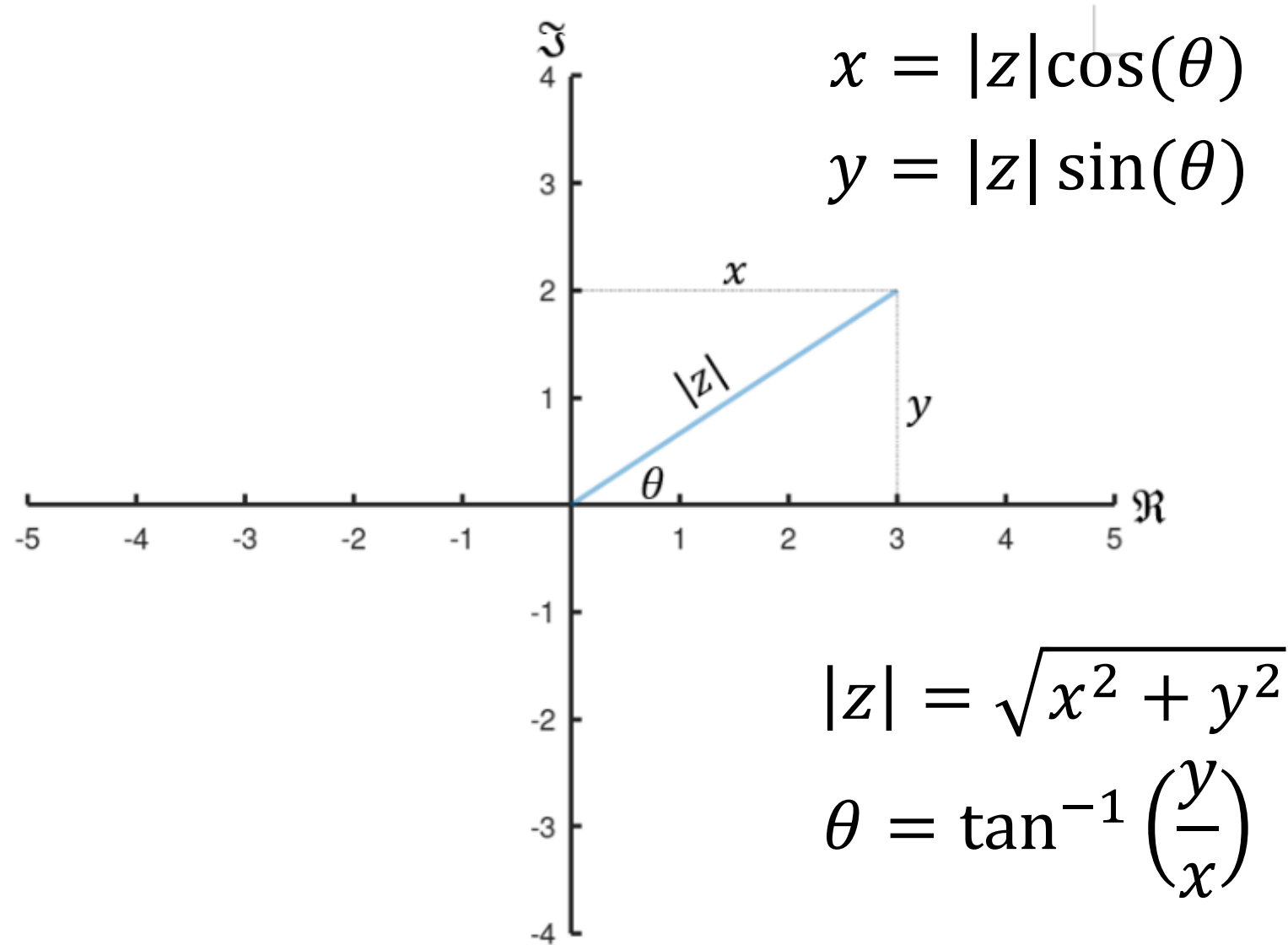
$$z = 12 \angle 60^\circ = 12 \angle \pi/3 = 12 e^{j\frac{\pi}{3}}$$



Konversi Rectangular $\leftrightarrow$ Polar



Konversi Rectangular $\leftrightarrow$ Polar



Conjugate (Bentuk Sekawan)

$$z = x + jy \rightarrow z^* = x - jy = |z|e^{-j\theta} = |z|\angle -\theta$$

Contoh :

$$\begin{aligned} z &= 5 + j3 \\ z^* &= 5 - j3 \end{aligned}$$



Penjumlahan

$$z_1 = x_1 + jy_1$$

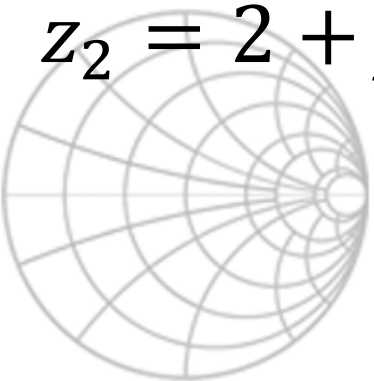
$$z_2 = x_2 + jy_2$$

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Contoh :

$$z_1 = 5 + j3$$

$$z_2 = 2 + j4$$



Penjumlahan

$$z_1 = x_1 + jy_1$$

$$z_2 = x_2 + jy_2$$

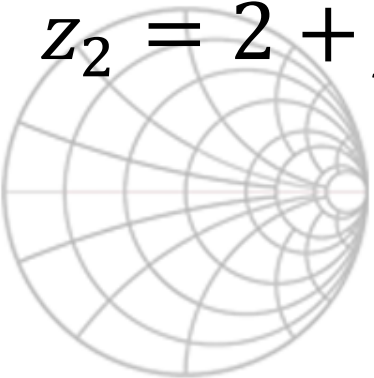
$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Contoh :

$$z_1 = 5 + j3$$

$$z_2 = 2 + j4$$

$$\begin{aligned} z_1 + z_2 &= (5 + 2) + j(3 + 4) \\ &= 7 + j7 \end{aligned}$$



Perkalian

$$z_1 = x_1 + jy_1$$

$$z_2 = x_2 + jy_2$$

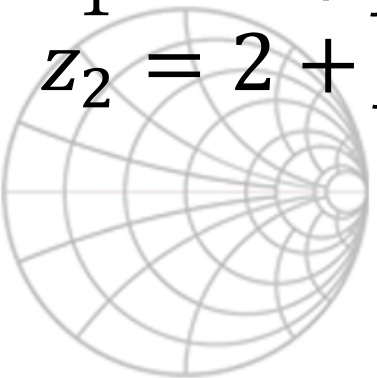
$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1) \end{aligned}$$

Contoh :

$$z_1 = 5 + j3$$

$$z_2 = 2 + j4$$

$$z_1 z_2 = (5 + j3)(2 + j4)$$

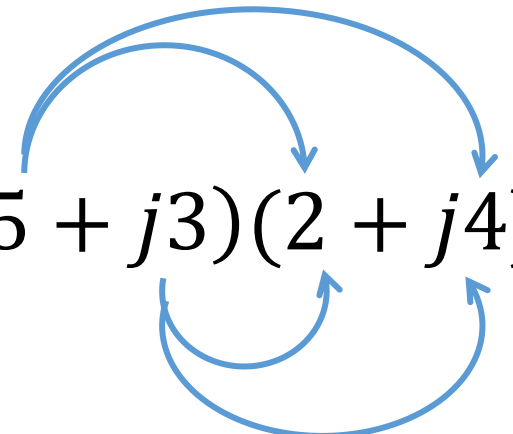


Perkalian

Contoh :

$$z_1 = 5 + j3$$

$$z_2 = 2 + j4$$

$$z_1 z_2 = (5 + j3)(2 + j4)$$


$$= (5)(2) + (5)(j4) + (j3)(2) + (j3)(j4)$$

$$= 10 + j20 + j6 - 12$$

$$= -2 + j26$$



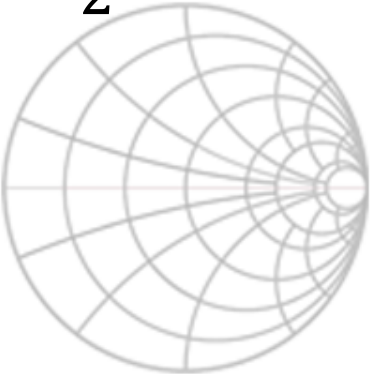
Perkalian

$$\begin{aligned} z_1 z_2 &= |z_1| e^{j\theta_1} \cdot |z_2| e^{j\theta_2} \\ &= |z_1| |z_2| e^{j(\theta_1 + \theta_2)} \\ &= |z_1| |z_2| (\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)) \end{aligned}$$

Contoh :

$$z_1 = 3 \angle 45^\circ$$

$$z_2 = 5 \angle 15^\circ$$

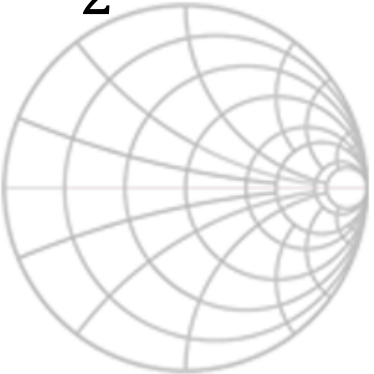


Perkalian

$$\begin{aligned}z_1 z_2 &= |z_1| e^{j\theta_1} \cdot |z_2| e^{j\theta_2} \\&= |z_1| |z_2| e^{j(\theta_1 + \theta_2)} \\&= |z_1| |z_2| (\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2))\end{aligned}$$

Contoh :

$$\begin{aligned}z_1 &= 3 \angle 45^\circ & z_1 z_2 &= (3)(5) \angle (45^\circ + 15^\circ) \\z_2 &= 5 \angle 15^\circ & &= 15 \angle (60^\circ)\end{aligned}$$



Perkalian dengan Conjugate

$$z_2 = x + jy \rightarrow z_2^* = x - jy$$

$$z_2 \times z_2^* = (x + jy)(x - jy)$$

$$\begin{aligned} & (x)(x) + (x)(-jy) + (jy)(x) + (jy)(-jy) \\ & x^2 - jxy + jxy + y^2 \\ & x^2 + y^2 \end{aligned}$$



Pembagian

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{x_1 + jy_1}{x_2 + jy_2} \\ &= \frac{(x_1 + jy_1)}{(x_2 + jy_2)} \cdot \frac{(x_2 - jy_2)}{(x_2 - jy_2)} \\ &= \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{(x_2^2 + y_2^2)}\end{aligned}$$

Contoh :

$$z_1 = 5 + j3 \quad z_2 = 2 + j4$$

$$\frac{z_1}{z_2} = \frac{5 + j3}{2 + j4}$$

Pembagian

Contoh :

$$z_1 = 5 + j3 \quad z_2 = 2 + j4$$

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{5 + j3}{2 + j4} \times \frac{2 - j4}{2 - j4} \\ &= \frac{(5)(2) + 5(-j4) + (j3)(2) + (j3)(-j4)}{2^2 + 4^2} \\ &= \frac{10 - j20 + j6 + 12}{20} \\ &= \frac{22 - j14}{20} \end{aligned}$$

Pembagian

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} \\ &= \frac{|z_1|}{|z_2|}e^{j(\theta_1-\theta_2)}\end{aligned}$$

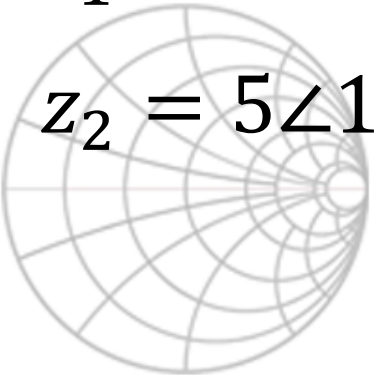
Contoh :

$$z_1 = 3\angle 45^\circ$$

$$z_1/z_2 = (3)/(5)\angle(45^\circ - 15^\circ)$$

$$z_2 = 5\angle 15^\circ$$

$$= \frac{3}{5}\angle(30^\circ)$$



Persamaan pada Bilangan Kompleks

- $-1 = e^{j\pi} = e^{-j\pi} = 1\angle 180^\circ$
- $j = e^{j\pi/2} = 1\angle 90^\circ$
- $-j = -e^{j\pi/2} = e^{-j\pi/2} = 1\angle -90^\circ$
- $\sqrt{j} = (e^{j\pi/2})^{1/2} = \pm e^{j\pi/4} = \frac{\pm(1+j)}{\sqrt{2}}$
- $\sqrt{-j} = \pm e^{-j\pi/4} = \frac{\pm(1-j)}{\sqrt{2}}$

